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MAHATMA GANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)REGULAREXAMINATIONMarch2026

FourthSEMESTER

DisciplineSpecificElective(DSE)-MG4DSEFSQ200-FOODPACKAGING TECHNOLOGY

(2024ADMISSIONONWARDS)

Duration:2hours **MaximumMarks:70**

PartA

MCQ

Answerallquestions

Eachquestioncarries1marks

1. Trays made from thermoformed plastic are termed?
Answer: a) Semi-rigid packaging

2. A screw closure is mainly used for which purpose?
Answer: b) Sealing containers

3. Dry lamination is preferred for food packaging because:
Answer: a) It provide strong and stable bonds

4. Select the difference between global migration and specific migration.
Answer: a) Specific migration measures individual chemicals

5. Hot melt lamination is often preferred over solvent lamination because it is:
Answer: b) Is faster and solvent free

6. Interpret which packaging mainly provides the first level of protection to food.
Answer: b) Primary

7. Main disadvantage of paper packaging is:
Answer: a) Moisture sensitivity

8. RFID stand for:
Answer: a) Radio Frequency Identification

9. Describe the principle of Gas Transmission Rate testing.
Answer: a) Measuring gas flow through a packaging material under controlled conditions

10. Select the scope of food packaging includes supporting export of food by:
Answer: c) Meeting international packaging standards

11. Polypropylene is widely used in food packaging because of its:
Answer: a) High melting point and chemical resistance

12. The ability of plastic resist deformation under load is known as?
Answer: c) Rigidity

13. The sealing method used for retort pouches:
Answer: c) Heat sealing

14. Select the packaging material property mainly assessed by puncture resistance:
Answer: b) Mechanical integrity

15. Define the purpose of protective lacquers inside food cans:
Answer: b) To prevent metal–food reaction

16. Identify the factor that does not affect WVTR:
Answer: d) Package color

17. Heat sealing property in laminates is mainly provided by:
Answer: b) Plastic layer

18. Oxygen is intentionally included in MAP for packaging of:
Answer: c) Fresh red meat

19. Global migration in food packaging refers to:
Answer: b) Total transfer of substances from packaging into food

20. List which role of packaging helps consumers make informed food choices:
Answer: a) Labelling information

PartB

ShortAnswer

Answerany6questions

Eachquestioncarries5marks

21. Explain the legal control of misleading claims on food packages.

Legal control of misleading claims on food packages is essential to protect consumers from false or deceptive information. Regulatory authorities such as food safety agencies establish strict guidelines for labeling and advertising. These laws ensure that claims related to nutrition, health benefits, ingredients, and quality are scientifically valid and clearly stated. Misleading terms like “natural,” “organic,” or “low-fat” must meet specific criteria before being used. Labels must not exaggerate benefits or hide important information such as allergens. Violations can result in penalties, product recalls, or legal action. Such regulations help maintain transparency, build consumer trust, and promote fair competition in the food industry.

22. Outline why amber and green glass are preferred for packaging light-sensitive food and beverage products.

Amber and green glass are commonly used for packaging light-sensitive food and beverages because they provide protection against harmful ultraviolet (UV) and visible light. Exposure to light can cause chemical changes in food products, leading to spoilage, loss of flavor, color degradation, and nutrient destruction. Amber glass offers the highest level of protection by blocking a significant portion of UV radiation, making it suitable for products like beer and certain oils. Green glass also provides moderate protection and is often used for beverages. By reducing light exposure, these colored glasses help maintain product quality, extend shelf life, and preserve sensory characteristics.

23. Write a short note on biodegradable plastics.

Biodegradable plastics are materials that can be broken down by microorganisms such as bacteria and fungi into natural substances like water, carbon dioxide, and biomass. These plastics are often made from renewable resources such as starch, cellulose, or polylactic acid (PLA). They are considered environmentally friendly alternatives to conventional plastics, as they reduce pollution and landfill accumulation. In food packaging, biodegradable plastics are used for trays, films, and containers. However, their effectiveness depends on proper disposal conditions, such as industrial composting facilities. While they offer sustainability benefits, challenges such as cost, durability, and limited shelf life still exist.

24. Discuss the role of tensile strength in handling and transportation of packaged foods.

Tensile strength refers to the ability of a packaging material to withstand pulling or stretching forces without breaking. It plays a crucial role in the handling and transportation of packaged foods, as packages are often subjected to stress during loading, unloading, and transit. Materials

with high tensile strength can resist tearing and damage, ensuring that the product remains protected. This property is especially important for flexible packaging such as plastic films and bags. Adequate tensile strength reduces the risk of package failure, product loss, and contamination. It also contributes to maintaining the integrity and safety of food products throughout the supply chain.

25. Describe tin-free steel (TFS) and mention its advantages over tinplate.

Tin-free steel (TFS), also known as electrolytic chromium-coated steel, is a type of steel sheet coated with chromium instead of tin. It is widely used in food packaging, especially for making cans and lids. TFS provides a smooth and attractive surface suitable for printing and coating. One of its main advantages over tinplate is its lower cost, as it does not use expensive tin. It also offers better resistance to corrosion and improved lacquer adhesion. However, unlike tinplate, TFS cannot be easily soldered and is usually welded. Overall, it is a cost-effective and efficient alternative for certain packaging applications.

26. Distinguish vibration and compression test used during transportation.

Vibration and compression tests are important methods used to evaluate the strength and durability of packaging during transportation. The vibration test simulates the conditions experienced during transit, such as movement and shaking, to assess how the package and its contents respond to continuous motion. It helps identify potential damage caused by loosening, shifting, or fatigue. On the other hand, the compression test measures the ability of a package to withstand stacking loads and external pressure. It evaluates the strength of packaging when multiple units are stacked during storage or transport. While vibration testing focuses on dynamic forces, compression testing deals with static load-bearing capacity.

27. Describe the pouches used in food packaging.

Pouches are flexible packaging solutions widely used in the food industry due to their convenience, lightweight nature, and cost-effectiveness. They are made from laminated materials such as plastic, aluminum foil, and paper, providing good barrier properties against moisture, oxygen, and light. Common types include stand-up pouches, flat pouches, retort pouches, and spouted pouches. These pouches are used for packaging snacks, beverages, ready-to-eat meals, and sauces. They are easy to handle, require less storage space, and often come with features like resealable zippers. Pouches enhance product shelf life while offering attractive presentation and consumer convenience.

28. Write a note on types of form-fill-seal packaging machines.

Form-fill-seal (FFS) packaging machines are automated systems used to form packages, fill them with products, and seal them in a continuous process. There are mainly two types: vertical form-fill-seal (VFFS) and horizontal form-fill-seal (HFFS) machines. VFFS machines form packages vertically and are commonly used for powders, grains, and snack foods. HFFS machines operate horizontally and are suitable for solid or semi-solid products such as biscuits and chocolates.

These machines improve efficiency, reduce labor costs, and ensure uniform packaging. They also maintain hygiene and are widely used in large-scale food processing industries.

PartC

Essay

Answerany**2**questions

Eachquestioncarries**10**marks

29. Explain bursting strength of packaging materials. Discuss the factors affecting it and its importance in food packaging applications.

Bursting strength is the ability of a packaging material, particularly paper and paperboard, to withstand internal pressure before it ruptures. It is measured using instruments like the Mullen tester, where pressure is applied until the material bursts. This property is crucial in determining the durability and resistance of packaging during handling, storage, and transportation. Several factors affect bursting strength, including the type and quality of raw materials, fiber length, bonding between fibers, thickness of the material, and moisture content. Higher moisture levels can weaken the structure, reducing strength. Manufacturing processes such as refining and pressing also influence fiber bonding and overall strength. The importance of bursting strength in food packaging lies in its ability to protect products from mechanical damage and external pressure. Strong packaging prevents leakage, contamination, and spoilage, ensuring product safety and maintaining quality throughout the supply chain.

30. Describe the manufacture of paper for packaging purpose and discuss how processing affects its properties.

The manufacture of paper for packaging involves several key steps, starting with the selection of raw materials such as wood pulp, recycled paper, or other fibrous materials. The process begins with pulping, where raw materials are broken down into fibers through mechanical or chemical methods. The pulp is then cleaned, screened, and sometimes bleached to improve quality. Next, the pulp is formed into sheets on a wire mesh, where water is drained, and fibers interlock to form a continuous web. This sheet is then pressed and dried to remove remaining moisture. Additional treatments such as coating, sizing, and calendaring may be applied to enhance properties. Processing conditions significantly affect paper properties such as strength, thickness, smoothness, porosity, and printability. For example, increased refining improves fiber bonding and strength, while excessive drying may reduce flexibility. Coating enhances barrier properties and appearance. Thus, careful control of processing parameters is essential to produce paper suitable for specific packaging applications.

31. Outline the protective, containment, and communication functions of food packaging in the food industry.

Food packaging performs three major functions: protection, containment, and communication. The protective function involves safeguarding food from physical, chemical, and biological

hazards such as moisture, oxygen, light, microorganisms, and mechanical damage. This helps in preserving quality, extending shelf life, and ensuring food safety. The containment function refers to holding the product in a convenient form for handling, storage, and transportation. It prevents leakage, spillage, and loss of product, making distribution efficient. The communication function provides essential information to consumers through labeling, including ingredients, nutritional facts, expiry date, usage instructions, and safety warnings. It also plays a role in marketing by attracting consumers through design and branding. Together, these functions ensure that food reaches consumers in a safe, convenient, and informative manner, supporting both industry operations and consumer needs.

32. Describe paper types, uses, and its applications.

Paper used in food packaging is available in various types, each designed for specific applications based on its properties. Common types include kraft paper, greaseproof paper, glassine paper, parchment paper, and paperboard. Kraft paper is strong and durable, widely used for bags and wrapping materials. Greaseproof paper resists oil and is used for packaging fatty foods like butter and bakery products. Glassine paper is smooth and translucent, often used for wrapping confectionery items. Parchment paper provides resistance to moisture and grease, making it suitable for baking and food wrapping. Paperboard, including corrugated board, is used for cartons, boxes, and bulk packaging. These materials are used in a wide range of applications such as food wrapping, takeaway containers, beverage cartons, and shipping boxes. Paper packaging is biodegradable, recyclable, and cost-effective, making it an environmentally friendly choice. Its versatility and adaptability make it an essential material in the food packaging industry.